(Please write your Enrollment Number)

MID-TERM EXAMINATION
(Course Name: B.Tech., IT/ECE/CSE) (Semester: 6th)
(February, 2023)OFF LINE mode

Subject Code: BIT 304	Subject: Cloud Computing
Time :1 ½Hours	Maximum Marks : 30
Note:Q. 1 is compulsory.	

Q1 Consider the following statement, **(2.5*4)**

"Virtualization aims to run multiple OS instances on a single server, whereas containerization runs a single OS instance, with multiple user spaces to isolate processes from one another."

In the context of below-mentioned technical terms, categorically differentiate between containerization and virtualization using suitable examples.

(a) Isolation
(b) Deployment
(c) Operating Systems
(d) Virtual Storage

Q2 (Attempt any Two Parts) **(5,5)**

Consider the following statement,

"Cloud-based hyperconnectivity advances business in Smart City and Industry 4.0."

(a) With reference to the hyperconnectivity in the cloud-enabled environment, suggest suitable Service Oriented Architecture for cloud-enabled IGDTUW Examination portal from Student's perspective.

(b) With reference to unique services and events in the IGDTUW Examination portal, design suitable Event Driven Architecture for cloud-enabled IGDTUW Examination portal from Faculty's perspective.

(c) With reference to hybrid cloud deployment model, design a suitable cloud architecture for IGDTUW ERP portal both from the frontend and backend perspective. Incorporate the scalability and the security compliance factors in the designed architecture for handling flexibility and infrastructure dependency issues.

Q3 (Attempt any Two Parts) **(5,5)**

Consider the following statement,

"De-parameterized areas in Jericho Forum Cloud Cube Model use both internal and external domains but the collaboration/sharing of data should not be seen as internal or external."

(a) With reference to Jericho Forum Cloud Cube Model, discuss the four-dimensional factors on the basis of which your cloud-enabled IGDTUW Examination portal would be categorized and hence, protected and secured.

(b) With reference to Platform-as-a-Service in Cloud Cube Model, discuss the significance of Vendor Lock-In problem and its Operational Limitations.

(c) With reference to Infrastructure-as-a-Service in Cloud Cube Model, discuss the significance of Multi-tenant Security and how would you ensure that Virtual Machines are adequately isolated within Multitenant cloud architecture.